Lecture 19 - Eigenvalues, Eigenvectors

$\det(A-\lambda I)=0$

$\text{Trace} = \lambda_1+\lambda_2+\cdots+\lambda_n$

Eigenvector: Ax parallel to x

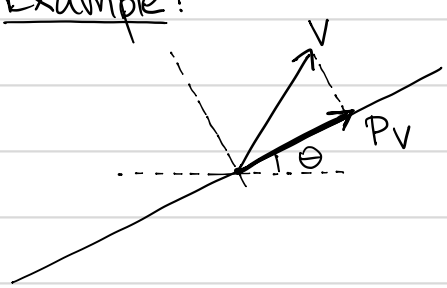
$$\boxed{Ax=\lambda x}$$

$x \neq 0$: eigenvector

λ: eigenvalue.

If A is singular, $\lambda=0$ is an eigenvalue.

Example:



$$P=\begin{bmatrix} c^2 & cs \\ cs & s^2 \end{bmatrix}$$

x on the line: $Px=x$ $\quad(\lambda=1)$

x perp. to line: $Px=0x$ $\quad(\lambda=0)$

Example: $A=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\qquad A\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$

$x=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\lambda=1$, $\quad x=\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $\lambda=1$

Fact:

$$\text{Sum of } \lambda\text{'s} = \underbrace{a_{11}+a_{22}+\cdots+a_{nn}}_{\text{trace}(A)}$$

Product of λ's $=\det(A)$.

How to Solve $Ax=\lambda x$:

Rewrite: $\underbrace{(A-\lambda I)}_{\text{singular}}x=0$ $\qquad (x\neq 0)$

$\rightarrow \det(A-\lambda I)=0$

$\rightarrow$ Find λ first!

Example: $A=\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

$$\det(A-\lambda I)=\begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix}=$$

$$=(3-\lambda)(3-\lambda)-1$$

$$=\lambda^2-6\lambda+8$$

$$=(\lambda-4)(\lambda-2)=0$$

$$\lambda_1=4, \quad \lambda_2=2.$$

$A-4I=\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ $\qquad x_1=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$A-2I=\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $\qquad x_2=\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

If $Ax = \lambda x$, then

$$(A+3I)x = \lambda x + 3x = (\lambda+3)x$$

<u>Warning</u>: $Ax = \lambda x \qquad By = \mu y$

$$\cancel{(A+B)z = (\lambda+\mu)z}$$

$$\cancel{ABz = \lambda\mu z}$$

<u>Example</u>: 90° rotation

$$Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \qquad \begin{aligned} \text{trace} &= 0+0 = \lambda_1 + \lambda_1 \\ \det &= 1 = \lambda_1 \lambda_2 \end{aligned}$$

$$\det(Q-\lambda I) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 = 0$$

$$\lambda_1 = i, \quad \lambda_2 = -i$$

<u>Example</u>: $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$

$$\det(A-\lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ 0 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 = 0$$

$$\lambda_1 = 3, \quad \lambda_2 = 3$$

$$(A-3I)x = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad x_2 = \cdots$$

No second independent eigenvector.